

02. Unsupervised Anomaly Detection (Isolation Forest)

본 노트북은 scikit-learn의 `IsolationForest` 모델을 활용하여 로그 데이터의 비지도 학습 이상치 탐지(Anomaly Detection)를 수행하고 최적의 `contamination` 비율을 튜닝하는 실습 공간입니다.

```
In [19]: %load_ext autoreload
          %autoreload 2

import os
import sys
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix, roc_auc_score

# 프로젝트 루트 경로를 sys.path에 추가하여 src 패키지를 찾을 수 있도록 함
sys.path.append(os.path.abspath(os.path.join("..")))

from src.parser import LogParser
from src.detector import AnomalyDetector

# 한글 폰트 설정 (추가됨)
import platform

if platform.system() == "Darwin": # Mac
    plt.rcParams["font.family"] = "AppleGothic"
elif platform.system() == "Windows": # Windows
    plt.rcParams["font.family"] = "Malgun Gothic"
plt.rcParams["axes.unicode_minus"] = False
```

The autoreload extension is already loaded. To reload it, use:
%reload_ext autoreload

1. 전처리 데이터 로드

```
In [20]: SSD_PATH = "/Volumes/T7/LogSentinel_Data/raw/"
          # CWD가 notebooks 디렉토리일 경우를 고려하여 프로젝트 루트 기준으로 상대경로를 절대경로로 변환함
          LOCAL_PATH = os.path.abspath(os.path.join("..", "data", "raw"))

def get_data_path(filename):
    ssd_file = os.path.join(SSD_PATH, filename)
    local_file = os.path.join(LOCAL_PATH, filename)
    if os.path.exists(ssd_file):
        return ssd_file
    return local_file
```

```
log_file_path = get_data_path("BGL_2k.log")

data = []
with open(log_file_path, "r", encoding="utf-8") as f:
    for line in f:
        parts = line.strip().split()
        if len(parts) < 10:
            continue
        label = parts[0]
        is_anomaly = 1 if label != "-" else 0
        message = " ".join(parts[9:])
        data.append({"is_anomaly": is_anomaly, "raw_message": message})

df = pd.DataFrame(data)
print(f"Loaded {len(df)} logs.")
```

Loaded 2000 logs.

In [21]: data

```

Out[21]: [{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': '63543 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': '162 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': '141 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': 'CE sym 2, at 0x0b85eee0, mask 0x05'},
{'is_anomaly': 1,
'raw_message': 'ciod: failed to read message prefix on control stream (Ci
oStream socket to 172.16.96.116:33569'},
{'is_anomaly': 1,
'raw_message': 'ciod: failed to read message prefix on control stream (Ci
oStream socket to 172.16.96.116:33370'},
{'is_anomaly': 0, 'raw_message': 'CE sym 20, at 0x1438f9e0, mask 0x40'},
{'is_anomaly': 0, 'raw_message': 'generating core.2275'},
{'is_anomaly': 0, 'raw_message': 'generating core.862'},
{'is_anomaly': 0, 'raw_message': 'generating core.728'},
{'is_anomaly': 0, 'raw_message': 'generating core.775'},
{'is_anomaly': 0, 'raw_message': 'generating core.3276'},
{'is_anomaly': 0, 'raw_message': 'generating core.1717'},
{'is_anomaly': 0, 'raw_message': 'generating core.3919'},
{'is_anomaly': 0, 'raw_message': 'generating core.2079'},
{'is_anomaly': 0, 'raw_message': 'generating core.1414'},
{'is_anomaly': 0, 'raw_message': 'generating core.3055'},
{'is_anomaly': 0, 'raw_message': 'generating core.201'},
{'is_anomaly': 0, 'raw_message': 'generating core.1125'},
{'is_anomaly': 0, 'raw_message': 'generating core.412'},
{'is_anomaly': 0, 'raw_message': 'generating core.7828'},
{'is_anomaly': 0, 'raw_message': 'generating core.5570'},
{'is_anomaly': 0, 'raw_message': 'generating core.8275'},
{'is_anomaly': 0, 'raw_message': 'generating core.4183'},
{'is_anomaly': 0, 'raw_message': 'generating core.6545'},
{'is_anomaly': 0, 'raw_message': 'generating core.4245'},
{'is_anomaly': 0, 'raw_message': 'generating core.6884'},
{'is_anomaly': 0,
'raw_message': 'force load/store alignment.....0'},
{'is_anomaly': 0, 'raw_message': 'generating core.6471'},
{'is_anomaly': 0, 'raw_message': 'generating core.4155'},
{'is_anomaly': 0, 'raw_message': 'generating core.449'},
{'is_anomaly': 0, 'raw_message': 'generating core.6990'},
{'is_anomaly': 0, 'raw_message': 'generating core.4876'},
{'is_anomaly': 0, 'raw_message': 'generating core.2218'},
{'is_anomaly': 0, 'raw_message': 'generating core.7518'},
{'is_anomaly': 0, 'raw_message': 'generating core.5854'},
{'is_anomaly': 0, 'raw_message': 'generating core.7457'},
{'is_anomaly': 0, 'raw_message': 'generating core.6896'},
{'is_anomaly': 0, 'raw_message': 'generating core.3488'},
{'is_anomaly': 0, 'raw_message': 'generating core.1172'},

```

```

{'is_anomaly': 0, 'raw_message': 'generating core.2286'},
{'is_anomaly': 0, 'raw_message': 'generating core.786'},
{'is_anomaly': 0, 'raw_message': 'generating core.2680'},
{'is_anomaly': 0, 'raw_message': 'generating core.1524'},
{'is_anomaly': 0, 'raw_message': 'generating core.4937'},
{'is_anomaly': 0, 'raw_message': 'generating core.6801'},
{'is_anomaly': 0, 'raw_message': 'generating core.4368'},
{'is_anomaly': 0, 'raw_message': 'generating core.851'},
{'is_anomaly': 0, 'raw_message': 'generating core.1744'},
{'is_anomaly': 0, 'raw_message': 'generating core.1588'},
{'is_anomaly': 0, 'raw_message': 'generating core.5307'},
{'is_anomaly': 0, 'raw_message': 'generating core.7192'},
{'is_anomaly': 0, 'raw_message': 'CE sym 20, at 0x01228120, mask 0x10'},
{'is_anomaly': 0, 'raw_message': 'CE sym 5, at 0x11042a80, mask 0x04'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': 'generating core.430'},
{'is_anomaly': 0, 'raw_message': 'generating core.174'},
{'is_anomaly': 0,
  'raw_message': 'ciod: cpu 0 at treeaddr 438 sent unrecognized message 0xf
ffffff'},
{'is_anomaly': 0, 'raw_message': 'generating core.976'},
{'is_anomaly': 0, 'raw_message': 'generating core.1990'},
{'is_anomaly': 0, 'raw_message': 'generating core.3638'},
{'is_anomaly': 0, 'raw_message': 'generating core.2690'},
{'is_anomaly': 0, 'raw_message': 'generating core.3804'},
{'is_anomaly': 0, 'raw_message': 'generating core.418'},
{'is_anomaly': 0, 'raw_message': 'generating core.310'},
{'is_anomaly': 0, 'raw_message': 'generating core.2409'},
{'is_anomaly': 0, 'raw_message': 'generating core.4182'},
{'is_anomaly': 0, 'raw_message': 'generating core.8153'},
{'is_anomaly': 0, 'raw_message': 'generating core.7836'},
{'is_anomaly': 0, 'raw_message': 'generating core.2911'},
{'is_anomaly': 0, 'raw_message': 'generating core.2921'},
{'is_anomaly': 0, 'raw_message': 'generating core.122'},
{'is_anomaly': 0, 'raw_message': 'generating core.1973'},
{'is_anomaly': 0,
  'raw_message': 'ciod: LOGIN chdir(/p/gb2/glosli/8M_5000K/t800) failed: No
such file or directory'},
{'is_anomaly': 0, 'raw_message': 'generating core.4984'},
{'is_anomaly': 0, 'raw_message': 'generating core.1822'},
{'is_anomaly': 0, 'raw_message': 'CE sym 28, at 0x110067e0, mask 0x02'},

```

[illegible]

[illegible]

```

{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 0,
  'raw_message': 'machine check: i-fetch.....0'},
{'is_anomaly': 0,
  'raw_message': 'machine check: i-fetch.....0'},
{'is_anomaly': 0,
  'raw_message': 'program interrupt: illegal instruction.....0'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 0,
  'raw_message': 'machine check: i-fetch.....0'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 1, 'raw_message': 'data storage interrupt'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x00004ed8'},
{'is_anomaly': 0, 'raw_message': 'exception syndrome register: 0x00800000'},
{'is_anomaly': 0, 'raw_message': 'exception syndrome register: 0x00800000'},
{'is_anomaly': 0, 'raw_message': 'exception syndrome register: 0x00800000'},
{'is_anomaly': 0, 'raw_message': 'exception syndrome register: 0x00800000'},
{'is_anomaly': 0,
  'raw_message': 'program interrupt: illegal instruction.....0'},
{'is_anomaly': 0,
  'raw_message': 'program interrupt: illegal instruction.....0'},
{'is_anomaly': 0, 'raw_message': 'data address: 0x00000002'},
{'is_anomaly': 0, 'raw_message': 'exception syndrome register: 0x00800000'},
{'is_anomaly': 0,
  'raw_message': 'machine check: i-fetch.....0'},
{'is_anomaly': 0,
  'raw_message': 'machine check: i-fetch.....0'},

```

```

    'raw_message': 'data store interrupt caused by dcbf.....0'},
    {'is_anomaly': 0,
     'raw_message': 'machine check: i-fetch.....0'},
    {'is_anomaly': 0,
     'raw_message': 'data store interrupt caused by dcbf.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: illegal instruction.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: illegal instruction.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: privileged instruction...0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: privileged instruction...0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: trap instruction.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: imprecise exception.....0'},
    {'is_anomaly': 0,
     'raw_message': 'store operation.....1'},
    {'is_anomaly': 0,
     'raw_message': 'store operation.....1'},
    {'is_anomaly': 0, 'raw_message': 'machine state register: 0x00002000'},
    {'is_anomaly': 0,
     'raw_message': 'data store interrupt caused by icbi.....0'},
    {'is_anomaly': 0, 'raw_message': 'problem state (0=sup,1=usr).....0'},
    {'is_anomaly': 0,
     'raw_message': 'data store interrupt caused by icbi.....0'},
    {'is_anomaly': 0,
     'raw_message': 'data store interrupt caused by icbi.....0'},
    {'is_anomaly': 0, 'raw_message': 'floating point instr. enabled.....1'},
    {'is_anomaly': 0, 'raw_message': 'floating point instr. enabled.....1'},
    {'is_anomaly': 0, 'raw_message': 'floating point instr. enabled.....1'},
    {'is_anomaly': 0,
     'raw_message': 'auxiliary processor.....0'},
    {'is_anomaly': 0, 'raw_message': 'machine check enable.....0'},
    {'is_anomaly': 0, 'raw_message': 'machine check enable.....0'},
    {'is_anomaly': 0, 'raw_message': 'floating pt ex mode 0 enable.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: unimplemented operation..0'},
    {'is_anomaly': 0, 'raw_message': 'debug interrupt enable.....0'},
    {'is_anomaly': 0, 'raw_message': 'floating pt ex mode 1 enable.....0'},
    {'is_anomaly': 0,
     'raw_message': 'byte ordering exception.....0'},
    {'is_anomaly': 0,
     'raw_message': 'byte ordering exception.....0'},
    {'is_anomaly': 0, 'raw_message': 'data address space.....0'},
    {'is_anomaly': 0, 'raw_message': 'data address space.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: imprecise exception.....0'},
    {'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: imprecise exception.....0'},
    {'is_anomaly': 0,
     'raw_message': 'program interrupt: fp cr update.....0'},
    {'is_anomaly': 0,

```



```
{'raw_message': 'guaranteed instruction cache block touch.0'},
{'is_anomaly': 0,
 'raw_message': 'guaranteed instruction cache block touch.0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: fp cr field .....0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: fp cr field .....0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: fp cr field .....0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: fp cr field .....0'},
{'is_anomaly': 0,
 'raw_message': 'guaranteed data cache block touch.....1'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: fp cr field .....0'},
{'is_anomaly': 0,
 'raw_message': 'force load/store alignment.....0'},
{'is_anomaly': 0,
 'raw_message': 'icache prefetch depth.....0'},
{'is_anomaly': 0, 'raw_message': 'machine state register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'machine state register: 0x00002000'},
{'is_anomaly': 0,
 'raw_message': 'icache prefetch threshold.....0'},
{'is_anomaly': 0,
 'raw_message': 'icache prefetch threshold.....0'},
{'is_anomaly': 0, 'raw_message': 'machine state register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'wait state enable.....0'},
{'is_anomaly': 0, 'raw_message': 'critical input interrupt enable...0'},
{'is_anomaly': 0, 'raw_message': 'problem state (0=sup,1=usr).....0'},
{'is_anomaly': 0, 'raw_message': 'problem state (0=sup,1=usr).....0'},
{'is_anomaly': 0, 'raw_message': 'floating point instr. enabled.....1'},
{'is_anomaly': 0, 'raw_message': 'special purpose registers:'},
{'is_anomaly': 0,
 'raw_message': 'lr:00004ed0 cr:28244842 xer:20000002 ctr:00086000'},
{'is_anomaly': 0, 'raw_message': 'rts internal error'},
{'is_anomaly': 0, 'raw_message': 'floating pt ex mode 0 enable.....0'},
{'is_anomaly': 0, 'raw_message': 'floating pt ex mode 0 enable.....0'},
{'is_anomaly': 0, 'raw_message': 'debug wait enable.....0'},
{'is_anomaly': 0, 'raw_message': 'floating pt ex mode 1 enable.....0'},
{'is_anomaly': 0, 'raw_message': 'instruction address space.....0'},
{'is_anomaly': 0, 'raw_message': 'data address space.....0'},
{'is_anomaly': 0, 'raw_message': 'data address space.....0'},
{'is_anomaly': 0, 'raw_message': 'data address space.....0'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'core configuration register: 0x00002000'},
{'is_anomaly': 0, 'raw_message': 'disable store gathering.....0'},
{'is_anomaly': 0,
```

```

    'raw_message': 'disable store gathering.....0'},
    {'is_anomaly': 0,
     'raw_message': 'disable store gathering.....0'},
    {'is_anomaly': 0, 'raw_message': 'generating core.42'},
    {'is_anomaly': 0,
     'raw_message': 'total of 1 ddr error(s) detected and corrected'},
    {'is_anomaly': 0, 'raw_message': 'generating core.145'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'generating core.1978'},
    {'is_anomaly': 0, 'raw_message': 'generating core.2040'},
    {'is_anomaly': 0, 'raw_message': 'generating core.15'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'generating core.16218'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
    {'is_anomaly': 0, 'raw_message': 'generating core.8531'},
    {'is_anomaly': 0, 'raw_message': 'generating core.2961'},
    {'is_anomaly': 0, 'raw_message': 'generating core.15966'},
    {'is_anomaly': 0, 'raw_message': 'generating core.42'},
    {'is_anomaly': 0, 'raw_message': 'generating core.93'},
    {'is_anomaly': 0, 'raw_message': 'generating core.2605'},
    {'is_anomaly': 0, 'raw_message': 'generating core.6037'},
    {'is_anomaly': 0, 'raw_message': 'generating core.886'},
    {'is_anomaly': 0, 'raw_message': 'generating core.3053'},
    {'is_anomaly': 0, 'raw_message': 'generating core.2790'},
    {'is_anomaly': 1,
     'raw_message': 'ciod: failed to read message prefix on control stream (Ci
oStream socket to 172.16.96.116:51706'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},

```

```

{'is_anomaly': 0, 'raw_message': 'generating core.1488'},
{'is_anomaly': 0, 'raw_message': 'CE sym 2, at 0x0b85f680, mask 0x0e'},
{'is_anomaly': 0, 'raw_message': 'generating core.2186'},
{'is_anomaly': 0, 'raw_message': 'generating core.978'},
{'is_anomaly': 0, 'raw_message': 'generating core.2808'},
{'is_anomaly': 0, 'raw_message': 'generating core.2486'},
{'is_anomaly': 0, 'raw_message': 'generating core.334'},
{'is_anomaly': 0, 'raw_message': 'generating core.5401'},
{'is_anomaly': 0, 'raw_message': 'generating core.14418'},
{'is_anomaly': 0, 'raw_message': 'generating core.2558'},
{'is_anomaly': 0, 'raw_message': 'generating core.9785'},
{'is_anomaly': 0, 'raw_message': 'generating core.4614'},
{'is_anomaly': 0, 'raw_message': 'generating core.7615'},
{'is_anomaly': 0, 'raw_message': 'generating core.15935'},
{'is_anomaly': 0,
  'raw_message': 'ciod: Error loading /home/draeger/testQboxhang-nozerobyte
bug-nosleepyescomm: invalid or missing program image, No such file or direc
tory'},
{'is_anomaly': 1,
  'raw_message': 'ciod: Error creating node map from file /p/gb2/cabot/mira
nda/newmaps/8k_128x64x1_8x4x4.map: No child processes'},
{'is_anomaly': 0, 'raw_message': 'CE sym 0, at 0x1b858280, mask 0x40'},
{'is_anomaly': 0, 'raw_message': 'generating core.8089'},
{'is_anomaly': 0, 'raw_message': 'generating core.4660'},
{'is_anomaly': 0, 'raw_message': 'generating core.721'},
{'is_anomaly': 0, 'raw_message': 'generating core.5287'},
{'is_anomaly': 0, 'raw_message': 'generating core.783'},
{'is_anomaly': 0,
  'raw_message': '1 ddr errors(s) detected and corrected on rank 0, symbol
17, bit 1'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0,
  'raw_message': 'ciod: Error loading ./runtime_malloc: invalid or missing
program image, No such file or directory'},
{'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},
{'is_anomaly': 0, 'raw_message': '199680 double-hummer alignment exception
s'},
{'is_anomaly': 0,
  'raw_message': '1969920 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '2576000 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '2576000 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '6182400 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '6182400 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '6182400 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '3091200 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '3091200 double-hummer alignment exceptions'},
{'is_anomaly': 0,

```

```

    'raw_message': '3091200 double-hummer alignment exceptions'},
    {'is_anomaly': 0, 'raw_message': 'CE sym 12, at 0x1b786f60, mask 0x04'},
    {'is_anomaly': 0, 'raw_message': 'CE sym 0, at 0x1b8584e0, mask 0x80'},
    {'is_anomaly': 0, 'raw_message': 'CE sym 23, at 0x03f89c00, mask 0x10'},
    {'is_anomaly': 0,
     'raw_message': '23431872 double-hummer alignment exceptions'},
    {'is_anomaly': 0,
     'raw_message': '259966605 double-hummer alignment exceptions'},
    {'is_anomaly': 0,
     'raw_message': '255244071 double-hummer alignment exceptions'},
    {'is_anomaly': 0,
     'raw_message': '259822751 double-hummer alignment exceptions'},
    {'is_anomaly': 0, 'raw_message': 'generating core.24721'},
    {'is_anomaly': 0, 'raw_message': 'generating core.4184'},
    {'is_anomaly': 0, 'raw_message': 'generating core.58109'},
    {'is_anomaly': 0, 'raw_message': 'generating core.9881'},
    {'is_anomaly': 0, 'raw_message': 'generating core.19579'},
    {'is_anomaly': 0, 'raw_message': 'generating core.4501'},
    {'is_anomaly': 0, 'raw_message': 'generating core.50230'},
    {'is_anomaly': 0, 'raw_message': 'generating core.62516'},
    {'is_anomaly': 0, 'raw_message': 'generating core.27921'},
    {'is_anomaly': 0, 'raw_message': 'generating core.16830'},
    {'is_anomaly': 0, 'raw_message': 'generating core.9369'},
    {'is_anomaly': 0, 'raw_message': 'generating core.10140'},
    {'is_anomaly': 0, 'raw_message': 'generating core.37749'},
    {'is_anomaly': 0, 'raw_message': 'generating core.9654'},
    {'is_anomaly': 0, 'raw_message': 'generating core.56155'},
    {'is_anomaly': 0, 'raw_message': 'generating core.40250'},
    {'is_anomaly': 0, 'raw_message': 'generating core.45353'},
    {'is_anomaly': 0, 'raw_message': 'generating core.9578'},
    {'is_anomaly': 0, 'raw_message': 'generating core.14856'},
    {'is_anomaly': 0, 'raw_message': 'generating core.62468'},
    {'is_anomaly': 0, 'raw_message': 'generating core.38415'},
    {'is_anomaly': 0, 'raw_message': 'generating core.39138'},
    {'is_anomaly': 0, 'raw_message': 'generating core.10394'},
    {'is_anomaly': 0, 'raw_message': 'generating core.41347'},
    {'is_anomaly': 0, 'raw_message': 'generating core.7619'},
    {'is_anomaly': 0, 'raw_message': 'generating core.54486'},
    {'is_anomaly': 0, 'raw_message': 'generating core.22421'},
    {'is_anomaly': 0, 'raw_message': 'generating core.18261'},
    {'is_anomaly': 0, 'raw_message': 'generating core.58616'},
    {'is_anomaly': 0, 'raw_message': 'generating core.47736'},
    {'is_anomaly': 0, 'raw_message': 'generating core.25048'},
    {'is_anomaly': 0, 'raw_message': 'generating core.58777'},
    {'is_anomaly': 0, 'raw_message': 'generating core.49137'},
    {'is_anomaly': 0, 'raw_message': 'generating core.42553'},
    {'is_anomaly': 0, 'raw_message': 'generating core.2109'},
    {'is_anomaly': 0, 'raw_message': 'generating core.718'},
    {'is_anomaly': 0, 'raw_message': 'generating core.10752'},
    {'is_anomaly': 0, 'raw_message': 'generating core.323'},
    {'is_anomaly': 0,
     'raw_message': 'ciod: Error loading /home/glosli/src/ddcMD/ddcMD/1.1.13/d
dcMdbglV: invalid or missing program image, No such file or directory'},
    {'is_anomaly': 0, 'raw_message': 'CE sym 5, at 0x11042a80, mask 0x04'},
    {'is_anomaly': 0, 'raw_message': 'instruction cache parity error correcte
d'},

```

```

{'is_anomaly': 0, 'raw_message': 'generating core.21112'},
{'is_anomaly': 0, 'raw_message': 'program interrupt'},
{'is_anomaly': 0, 'raw_message': 'generating core.28370'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: illegal instruction.....1'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: illegal instruction.....1'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: privileged instruction...0'},
{'is_anomaly': 0,
 'raw_message': 'data store interrupt caused by dcbf.....0'},
{'is_anomaly': 0,
 'raw_message': 'data store interrupt caused by icbi.....0'},
{'is_anomaly': 0,
 'raw_message': 'data store interrupt caused by icbi.....0'},
{'is_anomaly': 0,
 'raw_message': 'auxiliary processor.....0'},
{'is_anomaly': 0,
 'raw_message': 'auxiliary processor.....0'},
{'is_anomaly': 0,
 'raw_message': 'auxiliary processor.....0'},
{'is_anomaly': 0,
 'raw_message': 'auxiliary processor.....0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: unimplemented operation..0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: imprecise exception.....0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: imprecise exception.....0'},
{'is_anomaly': 0, 'raw_message': 'generating core.14700'},
{'is_anomaly': 0, 'raw_message': 'program interrupt'},
{'is_anomaly': 0, 'raw_message': 'program interrupt'},
{'is_anomaly': 0, 'raw_message': 'instruction address: 0x0062fe04'},
{'is_anomaly': 0, 'raw_message': 'exception syndrome register: 0x0800000
0'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: illegal instruction.....1'},
{'is_anomaly': 0,
 'raw_message': 'program interrupt: privileged instruction...0'},
{'is_anomaly': 0,
 'raw_message': 'data store interrupt caused by dcbf.....0'},
{'is_anomaly': 0,
 'raw_message': 'data store interrupt caused by dcbf.....0'},
{'is_anomaly': 0,
 'raw_message': 'auxiliary processor.....0'},
{'is_anomaly': 0, 'raw_message': 'CE sym 0, at 0x1b8594e0, mask 0x40'},
{'is_anomaly': 0, 'raw_message': 'Node card is not fully functional'},
{'is_anomaly': 0, 'raw_message': 'CE sym 10, at 0x08e70580, mask 0x08'},
{'is_anomaly': 0, 'raw_message': 'generating core.1786'},
{'is_anomaly': 0, 'raw_message': 'generating core.26918'},
{'is_anomaly': 0, 'raw_message': 'generating core.10125'},
{'is_anomaly': 0, 'raw_message': 'generating core.4230'},
{'is_anomaly': 0, 'raw_message': 'generating core.179'},
{'is_anomaly': 0, 'raw_message': 'generating core.1177'},
{'is_anomaly': 0, 'raw_message': 'generating core.32693'},
{'is_anomaly': 0, 'raw_message': 'generating core.7151'},
{'is_anomaly': 0, 'raw_message': 'generating core.7023'},
{'is_anomaly': 0, 'raw_message': '344040 double-hammer alignment exception

```

```

s'},
{'is_anomaly': 0, 'raw_message': '344040 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': '344040 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': 'generating core.29308'},
{'is_anomaly': 0, 'raw_message': 'generating core.2586'},
{'is_anomaly': 0, 'raw_message': 'generating core.6040'},
{'is_anomaly': 0, 'raw_message': 'generating core.16904'},
{'is_anomaly': 0, 'raw_message': 'generating core.26186'},
{'is_anomaly': 0, 'raw_message': 'generating core.26156'},
{'is_anomaly': 0, 'raw_message': 'generating core.2112'},
{'is_anomaly': 0, 'raw_message': 'generating core.23659'},
{'is_anomaly': 0, 'raw_message': 'generating core.1605'},
{'is_anomaly': 0, 'raw_message': 'generating core.16483'},
{'is_anomaly': 0, 'raw_message': 'generating core.257'},
{'is_anomaly': 0, 'raw_message': 'generating core.18751'},
{'is_anomaly': 0, 'raw_message': 'generating core.22783'},
{'is_anomaly': 0, 'raw_message': 'generating core.8021'},
{'is_anomaly': 0, 'raw_message': 'generating core.10734'},
{'is_anomaly': 0, 'raw_message': 'generating core.3613'},
{'is_anomaly': 0, 'raw_message': 'generating core.1916'},
{'is_anomaly': 0, 'raw_message': 'generating core.4246'},
{'is_anomaly': 0, 'raw_message': 'generating core.7777'},
{'is_anomaly': 0, 'raw_message': 'generating core.5935'},
{'is_anomaly': 0, 'raw_message': 'generating core.8691'},
{'is_anomaly': 0, 'raw_message': 'generating core.5512'},
{'is_anomaly': 0, 'raw_message': 'generating core.1123'},
{'is_anomaly': 0, 'raw_message': 'generating core.1641'},
{'is_anomaly': 0, 'raw_message': 'CE sym 0, at 0x1b858ca0, mask 0x40'},
{'is_anomaly': 0,
'raw_message': 'ciod: Error creating node map from file /p/gb2/welcome3/3
2k_128x256x1_8x4x4.map: Bad file descriptor'},
{'is_anomaly': 0, 'raw_message': 'CE sym 0, at 0x1b858280, mask 0x40'},
{'is_anomaly': 0, 'raw_message': 'CE sym 22, at 0x049af800, mask 0x02'},
{'is_anomaly': 0, 'raw_message': 'generating core.1165'},
{'is_anomaly': 0, 'raw_message': 'generating core.23615'},
{'is_anomaly': 0, 'raw_message': 'generating core.865'},
{'is_anomaly': 0, 'raw_message': 'generating core.1626'},
{'is_anomaly': 0, 'raw_message': 'generating core.16594'},
{'is_anomaly': 0, 'raw_message': 'generating core.30586'},
{'is_anomaly': 0, 'raw_message': 'generating core.18540'},
{'is_anomaly': 0, 'raw_message': 'generating core.9825'},
{'is_anomaly': 0, 'raw_message': 'generating core.6698'},
{'is_anomaly': 0, 'raw_message': 'generating core.4525'},
{'is_anomaly': 0, 'raw_message': 'generating core.28084'},
{'is_anomaly': 0, 'raw_message': 'generating core.6028'},
{'is_anomaly': 0, 'raw_message': 'generating core.536'},
{'is_anomaly': 0, 'raw_message': 'generating core.15655'},
{'is_anomaly': 0, 'raw_message': 'generating core.8628'},
{'is_anomaly': 0, 'raw_message': 'generating core.12122'},
{'is_anomaly': 0, 'raw_message': 'generating core.2210'},
{'is_anomaly': 0, 'raw_message': 'generating core.10312'},
{'is_anomaly': 0, 'raw_message': 'generating core.228'},
{'is_anomaly': 0, 'raw_message': 'generating core.332'},
{'is_anomaly': 0, 'raw_message': 'generating core.15740'},

```

[illegible]

```
{'is_anomaly': 0, 'raw_message': '458720 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': '458720 double-hummer alignment exception
s'},
{'is_anomaly': 0, 'raw_message': '458720 double-hummer alignment exception
s'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1146800 double-hummer alignment exceptions'},
{'is_anomaly': 0, 'raw_message': 'generating core.1114'},
{'is_anomaly': 0, 'raw_message': 'generating core.848'},
{'is_anomaly': 0, 'raw_message': 'generating core.2117'},
{'is_anomaly': 0, 'raw_message': 'generating core.3293'},
{'is_anomaly': 0, 'raw_message': 'generating core.1351'},
{'is_anomaly': 0, 'raw_message': 'generating core.17222'},
{'is_anomaly': 0, 'raw_message': 'generating core.20392'},
{'is_anomaly': 0, 'raw_message': 'generating core.31006'},
{'is_anomaly': 0, 'raw_message': 'generating core.20088'},
{'is_anomaly': 0, 'raw_message': 'generating core.26993'},
{'is_anomaly': 0, 'raw_message': 'generating core.20954'},
{'is_anomaly': 0, 'raw_message': 'generating core.6139'},
{'is_anomaly': 0, 'raw_message': 'generating core.10185'},
{'is_anomaly': 0, 'raw_message': 'generating core.13776'},
{'is_anomaly': 0, 'raw_message': 'generating core.19596'},
{'is_anomaly': 0, 'raw_message': 'generating core.15557'},
{'is_anomaly': 0, 'raw_message': 'generating core.4193'},
{'is_anomaly': 0, 'raw_message': 'generating core.7539'},
{'is_anomaly': 0, 'raw_message': 'generating core.22024'},
{'is_anomaly': 0, 'raw_message': 'generating core.22905'},
{'is_anomaly': 0, 'raw_message': 'generating core.24671'},
{'is_anomaly': 0, 'raw_message': 'generating core.17853'},
{'is_anomaly': 0, 'raw_message': 'generating core.25568'},
{'is_anomaly': 0, 'raw_message': 'generating core.23080'},
{'is_anomaly': 0, 'raw_message': 'generating core.9739'},
{'is_anomaly': 0, 'raw_message': 'generating core.7187'},
{'is_anomaly': 0, 'raw_message': 'generating core.23438'},
{'is_anomaly': 0, 'raw_message': 'generating core.7881'},
{'is_anomaly': 0, 'raw_message': 'generating core.9778'},
{'is_anomaly': 0, 'raw_message': 'generating core.7915'},
{'is_anomaly': 0, 'raw_message': 'generating core.3006'},
{'is_anomaly': 0, 'raw_message': 'generating core.14627'},
{'is_anomaly': 0, 'raw_message': 'generating core.2348'},
{'is_anomaly': 0, 'raw_message': 'generating core.226'},
{'is_anomaly': 0, 'raw_message': 'generating core.1219'},
{'is_anomaly': 0, 'raw_message': 'generating core.8877'},
```



```

{'is_anomaly': 0, 'raw_message': 'generating core.13558'},
{'is_anomaly': 0, 'raw_message': 'generating core.404'},
{'is_anomaly': 0,
 'raw_message': 'ciod: Missing or invalid fields on line 1 of node map fil
e /home/auselton/bgl/64mps.sequential.mapfile'},
{'is_anomaly': 0,
 'raw_message': '1 torus receiver z+ input pipe error(s) (dcr 0x02f0) dete
cted and corrected'},
{'is_anomaly': 0, 'raw_message': 'generating core.631'},
{'is_anomaly': 0, 'raw_message': 'generating core.998'},
{'is_anomaly': 0, 'raw_message': 'generating core.1129'},
{'is_anomaly': 0, 'raw_message': 'generating core.424'},
{'is_anomaly': 0, 'raw_message': 'generating core.1213'},
{'is_anomaly': 0, 'raw_message': 'generating core.1433'},
{'is_anomaly': 0, 'raw_message': 'generating core.2950'},
{'is_anomaly': 0, 'raw_message': 'generating core.319'},
{'is_anomaly': 0, 'raw_message': 'generating core.303'},
{'is_anomaly': 0, 'raw_message': 'generating core.1998'},
{'is_anomaly': 0,
 'raw_message': '632431728 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '634368144 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '632254896 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '633283776 double-hummer alignment exceptions'},
{'is_anomaly': 0, 'raw_message': 'generating core.841'},
{'is_anomaly': 0, 'raw_message': 'generating core.815'},
{'is_anomaly': 0, 'raw_message': 'generating core.57762'},
{'is_anomaly': 0, 'raw_message': 'generating core.3436'},
{'is_anomaly': 0, 'raw_message': 'generating core.4592'},
{'is_anomaly': 0, 'raw_message': 'generating core.149'},
{'is_anomaly': 0, 'raw_message': 'generating core.62507'},
{'is_anomaly': 0, 'raw_message': 'generating core.1807'},
{'is_anomaly': 0, 'raw_message': 'Node card is not fully functional'},
{'is_anomaly': 0,
 'raw_message': 'ciod: duplicate canonical-rank 31 to logical-rank 0 mappi
ng at line 3 of node map file /p/gb2/pakin1/sweep3d-5x5x400-10mk-3mmi-1024p
es-sweep/sweep.map'},
{'is_anomaly': 0,
 'raw_message': 'ciod: Error creating node map from file /p/gb2/pakin1/swe
ep3d-5x5x400-10mk-3mmi-1024pes-xyzt/xyzt.map: Block device required'},
{'is_anomaly': 0, 'raw_message': 'generating core.9652'},
{'is_anomaly': 0, 'raw_message': 'generating core.11154'},
{'is_anomaly': 0, 'raw_message': 'generating core.7571'},
{'is_anomaly': 0, 'raw_message': 'generating core.12364'},
{'is_anomaly': 0, 'raw_message': 'generating core.13817'},
{'is_anomaly': 0, 'raw_message': 'generating core.14349'},
{'is_anomaly': 0, 'raw_message': 'generating core.8897'},
{'is_anomaly': 0, 'raw_message': 'generating core.7946'},
{'is_anomaly': 0, 'raw_message': 'generating core.10299'},
{'is_anomaly': 0, 'raw_message': 'generating core.9949'},
{'is_anomaly': 0, 'raw_message': 'generating core.12012'},
{'is_anomaly': 0, 'raw_message': 'generating core.9051'},
{'is_anomaly': 0, 'raw_message': 'generating core.7752'},
{'is_anomaly': 0, 'raw_message': 'generating core.10793'},

```

```
{ 'is_anomaly': 0, 'raw_message': 'generating core.24671'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.40979'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.18129'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.47738'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.42109'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.6682'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.7672'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.47122'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.17493'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.23664'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.19296'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.6553'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.56320'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.3824'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5781'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.55191'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.38077'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.41557'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.36539'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.51900'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.33018'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.29433'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.36248'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.41080'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.57878'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.245'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2315'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.30603'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.17517'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2633'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.50240'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.10211'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.57863'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.51808'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.59907'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.25382'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.8646'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.7167'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14589'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2300'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14652'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14807'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.31758'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.7205'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.10247'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.30230'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2299'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.28666'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2481'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.52376'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.10930'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.51134'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5235'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.50841'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.48433'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.34955'},
```

```
{ 'is_anomaly': 0, 'raw_message': 'generating core.46408'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.35872'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.133'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.4622'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.58351'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.9158'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.22376'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.12329'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.4035'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.15459'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.55384'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5342'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.46559'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.25604'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.61172'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.25271'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.34015'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.63070'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.31192'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.40344'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.62462'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.40528'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.42234'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.23259'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.24314'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.4180'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.24538'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.26078'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.30706'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.8147'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.57391'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.11784'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.61963'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.33163'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.30275'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.15753'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.7362'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.8575'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.9784'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.11921'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.11879'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.13067'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.13131'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14071'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.8544'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.8531'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.13946'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.16095'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.7183'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.1988'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2347'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2790'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.2573'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.3985'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.13592'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.10019'},
```

```
{'is_anomaly': 0, 'raw_message': 'generating core.10842'},
{'is_anomaly': 0, 'raw_message': 'generating core.8138'},
{'is_anomaly': 0,
 'raw_message': '1365301360 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1315337824 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '1401390016 double-hummer alignment exceptions'},
{'is_anomaly': 0, 'raw_message': 'generating core.32657'},
{'is_anomaly': 0, 'raw_message': 'generating core.26835'},
{'is_anomaly': 0, 'raw_message': 'generating core.28242'},
{'is_anomaly': 0, 'raw_message': 'generating core.20970'},
{'is_anomaly': 0, 'raw_message': 'generating core.18514'},
{'is_anomaly': 0, 'raw_message': 'generating core.19321'},
{'is_anomaly': 0, 'raw_message': 'generating core.13753'},
{'is_anomaly': 0, 'raw_message': 'generating core.23542'},
{'is_anomaly': 0, 'raw_message': 'generating core.28918'},
{'is_anomaly': 0, 'raw_message': 'generating core.21145'},
{'is_anomaly': 0, 'raw_message': 'generating core.8832'},
{'is_anomaly': 0, 'raw_message': 'generating core.20015'},
{'is_anomaly': 0, 'raw_message': 'generating core.11564'},
{'is_anomaly': 0, 'raw_message': 'generating core.1557'},
{'is_anomaly': 0, 'raw_message': 'generating core.5167'},
{'is_anomaly': 0, 'raw_message': 'generating core.3111'},
{'is_anomaly': 0, 'raw_message': 'generating core.7193'},
{'is_anomaly': 0, 'raw_message': 'generating core.5861'},
{'is_anomaly': 0, 'raw_message': 'generating core.15425'},
{'is_anomaly': 0, 'raw_message': 'generating core.560'},
{'is_anomaly': 0, 'raw_message': 'generating core.5760'},
{'is_anomaly': 0, 'raw_message': 'generating core.13752'},
{'is_anomaly': 0, 'raw_message': 'generating core.10860'},
{'is_anomaly': 0, 'raw_message': 'generating core.12536'},
{'is_anomaly': 0, 'raw_message': 'generating core.19744'},
{'is_anomaly': 0, 'raw_message': 'generating core.8803'},
{'is_anomaly': 0, 'raw_message': 'generating core.28238'},
{'is_anomaly': 0, 'raw_message': 'generating core.12643'},
{'is_anomaly': 0, 'raw_message': 'generating core.12927'},
{'is_anomaly': 0, 'raw_message': 'generating core.6917'},
{'is_anomaly': 0, 'raw_message': 'generating core.21353'},
{'is_anomaly': 0, 'raw_message': 'generating core.17276'},
{'is_anomaly': 0, 'raw_message': 'generating core.26955'},
{'is_anomaly': 0, 'raw_message': 'generating core.11135'},
{'is_anomaly': 0, 'raw_message': 'generating core.6711'},
{'is_anomaly': 0, 'raw_message': 'generating core.22283'},
{'is_anomaly': 0, 'raw_message': 'generating core.21175'},
{'is_anomaly': 0, 'raw_message': 'generating core.28329'},
{'is_anomaly': 0, 'raw_message': 'generating core.20367'},
{'is_anomaly': 0, 'raw_message': 'generating core.23838'},
{'is_anomaly': 0, 'raw_message': 'generating core.16300'},
{'is_anomaly': 0, 'raw_message': 'generating core.7623'},
{'is_anomaly': 0, 'raw_message': 'generating core.1957'},
{'is_anomaly': 0, 'raw_message': 'generating core.3292'},
{'is_anomaly': 0, 'raw_message': 'generating core.3249'},
{'is_anomaly': 0, 'raw_message': 'generating core.2606'},
{'is_anomaly': 0, 'raw_message': 'generating core.2233'},
{'is_anomaly': 0, 'raw_message': 'generating core.3959'},
```

```

{'is_anomaly': 0, 'raw_message': 'generating core.2880'},
{'is_anomaly': 0, 'raw_message': 'generating core.3792'},
{'is_anomaly': 0, 'raw_message': 'generating core.2971'},
{'is_anomaly': 0, 'raw_message': 'generating core.3269'},
{'is_anomaly': 0, 'raw_message': 'generating core.24376'},
{'is_anomaly': 0, 'raw_message': 'generating core.31435'},
{'is_anomaly': 0, 'raw_message': 'generating core.3851'},
{'is_anomaly': 0, 'raw_message': 'generating core.20257'},
{'is_anomaly': 0, 'raw_message': 'generating core.32257'},
{'is_anomaly': 0, 'raw_message': 'generating core.7068'},
{'is_anomaly': 0, 'raw_message': 'generating core.4329'},
{'is_anomaly': 0, 'raw_message': 'generating core.16600'},
{'is_anomaly': 0, 'raw_message': 'generating core.6519'},
{'is_anomaly': 0, 'raw_message': 'generating core.15313'},
{'is_anomaly': 0, 'raw_message': 'generating core.16623'},
{'is_anomaly': 0, 'raw_message': 'generating core.10429'},
{'is_anomaly': 0, 'raw_message': 'generating core.32116'},
{'is_anomaly': 0, 'raw_message': 'generating core.15924'},
{'is_anomaly': 0, 'raw_message': 'CE sym 2, at 0x0d09f1c0, mask 0x80'},
{'is_anomaly': 0,
  'raw_message': 'ciod: Error loading /home/greeno/raptor.new.dev.3d.BGL.CX
X_XL.MPI.ex: invalid or missing program image, No such file or directory'},
{'is_anomaly': 1, 'raw_message': 'rts panic! - stopping execution'},
{'is_anomaly': 1, 'raw_message': 'rts panic! - stopping execution'},
{'is_anomaly': 0, 'raw_message': 'generating core.9079'},
{'is_anomaly': 0, 'raw_message': 'generating core.6425'},
{'is_anomaly': 0, 'raw_message': 'generating core.3400'},
{'is_anomaly': 0, 'raw_message': 'generating core.6711'},
{'is_anomaly': 0, 'raw_message': 'generating core.2263'},
{'is_anomaly': 0, 'raw_message': 'generating core.9386'},
{'is_anomaly': 0, 'raw_message': 'generating core.4061'},
{'is_anomaly': 0, 'raw_message': 'generating core.14070'},
{'is_anomaly': 0, 'raw_message': 'generating core.14021'},
{'is_anomaly': 0, 'raw_message': 'generating core.1715'},
{'is_anomaly': 0, 'raw_message': 'generating core.4489'},
{'is_anomaly': 0, 'raw_message': 'generating core.2092'},
{'is_anomaly': 0, 'raw_message': 'generating core.5967'},
{'is_anomaly': 0,
  'raw_message': 'total of 1 ddr error(s) detected and corrected'},
{'is_anomaly': 0,
  'raw_message': '118661512 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '112099344 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '113170696 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '117400328 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '112762744 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '116130784 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '114147592 double-hummer alignment exceptions'},
{'is_anomaly': 0,
  'raw_message': '113622512 double-hummer alignment exceptions'},
{'is_anomaly': 0,

```

```
'raw_message': '114059360 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '114091808 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '115325776 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '113982272 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '113355952 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '115193048 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '111990296 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '112625368 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '116249832 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '112481672 double-hummer alignment exceptions'},
{'is_anomaly': 0,
 'raw_message': '116332728 double-hummer alignment exceptions'},
{'is_anomaly': 0, 'raw_message': 'generating core.691'},
{'is_anomaly': 0, 'raw_message': 'generating core.52403'},
{'is_anomaly': 0, 'raw_message': 'generating core.12883'},
{'is_anomaly': 0, 'raw_message': 'generating core.4159'},
{'is_anomaly': 0, 'raw_message': 'generating core.5054'},
{'is_anomaly': 0, 'raw_message': 'generating core.17150'},
{'is_anomaly': 0, 'raw_message': 'generating core.47314'},
{'is_anomaly': 0, 'raw_message': 'generating core.5520'},
{'is_anomaly': 0, 'raw_message': 'generating core.17817'},
{'is_anomaly': 0, 'raw_message': 'generating core.34898'},
{'is_anomaly': 0, 'raw_message': 'generating core.47007'},
{'is_anomaly': 0, 'raw_message': 'generating core.18363'},
{'is_anomaly': 0, 'raw_message': 'generating core.22494'},
{'is_anomaly': 0, 'raw_message': 'CE sym 16, at 0x0456cd40, mask 0x04'},
{'is_anomaly': 0, 'raw_message': 'generating core.26059'},
{'is_anomaly': 0, 'raw_message': 'generating core.17999'},
{'is_anomaly': 0, 'raw_message': 'generating core.30671'},
{'is_anomaly': 0, 'raw_message': 'generating core.2600'},
{'is_anomaly': 0, 'raw_message': 'generating core.15192'},
{'is_anomaly': 0, 'raw_message': 'generating core.26224'},
{'is_anomaly': 0, 'raw_message': 'generating core.25712'},
{'is_anomaly': 0, 'raw_message': 'generating core.22807'},
{'is_anomaly': 0, 'raw_message': 'generating core.284'},
{'is_anomaly': 0, 'raw_message': 'generating core.5381'},
{'is_anomaly': 0, 'raw_message': 'generating core.27051'},
{'is_anomaly': 0, 'raw_message': 'generating core.29420'},
{'is_anomaly': 0, 'raw_message': 'generating core.8374'},
{'is_anomaly': 0, 'raw_message': 'generating core.15966'},
{'is_anomaly': 0, 'raw_message': 'generating core.7855'},
{'is_anomaly': 0, 'raw_message': 'generating core.2296'},
{'is_anomaly': 0, 'raw_message': 'generating core.1103'},
{'is_anomaly': 0, 'raw_message': 'generating core.2010'},
{'is_anomaly': 0, 'raw_message': 'generating core.3181'},
{'is_anomaly': 0, 'raw_message': 'generating core.32010'},
{'is_anomaly': 0, 'raw_message': 'generating core.9651'},
```

```
{ 'is_anomaly': 0, 'raw_message': 'generating core.10057'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5939'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.12984'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.17467'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.29367'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14852'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.4862'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.27323'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.26646'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5428'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.12212'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.3343'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.19942'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.18324'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.3510'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.17851'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.30632'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.26735'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.18689'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.27698'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.10645'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5009'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5785'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.4505'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.27577'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.32755'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.30675'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.11493'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.5671'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.27926'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.15967'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.24172'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.29732'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.17526'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.19481'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.23873'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14603'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.15737'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.23119'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.14824'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.15648'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.7156'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.22967'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.3980'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.11473'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.13043'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.28011'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.27197'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.27704'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.9337'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.22886'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.22325'},
{ 'is_anomaly': 0, 'raw_message': 'generating core.23998'},
{ 'is_anomaly': 0,
  'raw_message': 'ciod: Error creating node map from file /home/pakin1/swee
p3d-2.2b/results/random1-8x32x32x2.map: Permission denied'},
```

```

{'is_anomaly': 0,
 'raw_message': 'ciod: Z coordinate 32 exceeds physical dimension 32 at line 33 of node map file /p/gb2/pakin1/contention-32768cpes-torus/xyzt-1x1x32768x1.map'},
{'is_anomaly': 0, 'raw_message': 'generating core.4615'},
{'is_anomaly': 0, 'raw_message': 'generating core.1682'},
{'is_anomaly': 0, 'raw_message': 'generating core.896'},
{'is_anomaly': 0, 'raw_message': 'generating core.4675'},
{'is_anomaly': 0, 'raw_message': 'generating core.84'},
{'is_anomaly': 0, 'raw_message': 'generating core.1227'},
{'is_anomaly': 0,
 'raw_message': '1 ddr errors(s) detected and corrected on rank 0, symbol 33, bit 7'},
{'is_anomaly': 0, 'raw_message': 'generating core.59955'},
{'is_anomaly': 0, 'raw_message': 'generating core.38143'},
{'is_anomaly': 0, 'raw_message': 'generating core.41718'},
{'is_anomaly': 0, 'raw_message': 'generating core.29431'},
{'is_anomaly': 0, 'raw_message': 'generating core.33399'},
{'is_anomaly': 0, 'raw_message': 'generating core.11379'},
{'is_anomaly': 0, 'raw_message': 'generating core.36339'},
{'is_anomaly': 0, 'raw_message': 'generating core.42833'},
{'is_anomaly': 0, 'raw_message': 'generating core.20595'},
{'is_anomaly': 0, 'raw_message': 'generating core.17560'},
{'is_anomaly': 0, 'raw_message': 'generating core.17212'},
{'is_anomaly': 0, 'raw_message': 'generating core.26131'},
{'is_anomaly': 0, 'raw_message': 'generating core.45777'},
{'is_anomaly': 0, 'raw_message': 'generating core.62940'},
{'is_anomaly': 0, 'raw_message': 'generating core.27867'},
{'is_anomaly': 0, 'raw_message': 'generating core.21016'},
{'is_anomaly': 0, 'raw_message': 'generating core.465'},
{'is_anomaly': 0, 'raw_message': 'generating core.55352'},
{'is_anomaly': 0, 'raw_message': 'generating core.44723'},
{'is_anomaly': 0, 'raw_message': 'generating core.36787'},
{'is_anomaly': 0, 'raw_message': 'generating core.29903'},
{'is_anomaly': 0, 'raw_message': 'generating core.33706'},
{'is_anomaly': 0, 'raw_message': 'generating core.8879'},
{'is_anomaly': 0, 'raw_message': 'generating core.50505'},
{'is_anomaly': 0, 'raw_message': 'generating core.29899'},
{'is_anomaly': 0, 'raw_message': 'generating core.46536'},
{'is_anomaly': 0, 'raw_message': 'generating core.50988'},
{'is_anomaly': 0, 'raw_message': 'generating core.22273'},
{'is_anomaly': 0, 'raw_message': 'generating core.46157'},
{'is_anomaly': 0, 'raw_message': 'generating core.59042'},
{'is_anomaly': 0, 'raw_message': 'generating core.29543'},
{'is_anomaly': 0, 'raw_message': 'generating core.22119'},
{'is_anomaly': 0, 'raw_message': 'generating core.4900'},
{'is_anomaly': 0, 'raw_message': 'generating core.30256'},
{'is_anomaly': 0, 'raw_message': 'generating core.64146'},
{'is_anomaly': 0, 'raw_message': 'generating core.58328'},
{'is_anomaly': 0, 'raw_message': 'generating core.20241'},
{'is_anomaly': 0, 'raw_message': 'generating core.24337'},
{'is_anomaly': 0, 'raw_message': 'generating core.21045'},
{'is_anomaly': 0, 'raw_message': 'generating core.47256'},
{'is_anomaly': 0, 'raw_message': 'generating core.45183'},
{'is_anomaly': 0, 'raw_message': 'generating core.46654'},
{'is_anomaly': 0, 'raw_message': 'generating core.21374'},

```



```
{'is_anomaly': 0, 'raw_message': 'generating core.45880'},
{'is_anomaly': 0, 'raw_message': 'generating core.20691'},
{'is_anomaly': 0, 'raw_message': 'generating core.45943'},
...]
```

2. 파싱 및 TF-IDF 수치화

```
In [22]: parser = LogParser(max_features=500)
df["cleaned_message"] = df["raw_message"].apply(parser.clean_log)
X = parser.fit_transform(df["cleaned_message"])
y = df["is_anomaly"].values
```

```
In [23]: X.shape
```

```
Out[23]: (2000, 500)
```

```
In [24]: X
```

```
Out[24]: array([[0., 0., 0., ..., 0., 0., 0.],
                [0., 0., 0., ..., 0., 0., 0.],
                [0., 0., 0., ..., 0., 0., 0.],
                ...,
                [0., 0., 0., ..., 0., 0., 0.],
                [0., 0., 0., ..., 0., 0., 0.],
                [0., 0., 0., ..., 0., 0., 0.]])
```

```
In [25]: y.shape
```

```
Out[25]: (2000,)
```

```
In [26]: y
```

```
Out[26]: array([0, 0, 0, ..., 0, 0, 0])
```

3. Isolation Forest 학습 및 오염도(contamination) 하이퍼 파라미터 스캔

```
In [27]: # 경로 설정 및 방어 코딩 (T7 SSD 장착 여부 확인)
SSD_MODELS_PATH = "/Volumes/T7/LogSentinel_Data/models"
LOCAL_MODELS_PATH = os.path.abspath(os.path.join(".", "models"))
ANOMAL_DETECT_EVAL_FILE = 'confusion_matrix_and_rocauc_2k_per_c.txt'

# T7 SSD 디렉토리가 존재하는지 확인
if os.path.exists("/Volumes/T7"):
    models_dir = SSD_MODELS_PATH
else:
    models_dir = LOCAL_MODELS_PATH

os.makedirs(models_dir, exist_ok=True)
anomal_detection_eval_file = os.path.join(models_dir, ANOMAL_DETECT_EVAL_FILE)

contamination_ratios = [0.01, 0.05, 0.075, 0.1, 0.15]
```

```

# 파일에 저장할 데이터 헤더 작성 및 쓰기 준비
try:
    with open(anomal_detection_eval_file, "w", encoding="utf-8") as f:
        f.write("contamination,tn,fp,fn,tp,roc_auc\n") # Header 행
        print(f"Metrics file initialized at: {anomal_detection_eval_file}")
except Exception as e:
    print(f"Failed to initialize metrics file at {anomal_detection_eval_file}")

for c in contamination_ratios:
    detector = AnomalyDetector(contamination=c)
    detector.train(X)
    preds = detector.predict(X)
    pred_anomaly = (preds == -1).astype(int)

    print(f"\n[Contamination: {c}]")
    tn, fp, fn, tp = confusion_matrix(y, pred_anomaly).ravel()
    print(f"Confusion Matrix: TN={tn}, FP={fp}, FN={fn}, TP={tp}")
    # ROC AUC Score
    scores = -detector.get_anomaly_scores(X) # 높은 값이 아노말리를 나타내도록 부호 반전
    roc_auc = roc_auc_score(y, scores)
    print(f"ROC AUC Score: {roc_auc:.4f}")

    # 구분자(,)를 포함하여 파일에 추가 저장
    try:
        with open(anomal_detection_eval_file, "a", encoding="utf-8") as f:
            f.write(f"{c},{tn},{fp},{fn},{tp},{roc_auc:.4f}\n") # Data 행들
    except Exception as e:
        print(f"Failed to append metric for contamination {c} to file: {e}")

```

Metrics file initialized at: /Users/joshuapark/Desktop/Dev/soromiso/LogSentinel/models/confusion_matrix_and_rocauc_2k_per_c.txt

[Contamination: 0.01]

Confusion Matrix: TN=1857, FP=0, FN=143, TP=0

ROC AUC Score: 0.6525

[Contamination: 0.05]

Confusion Matrix: TN=1762, FP=95, FN=140, TP=3

ROC AUC Score: 0.6525

[Contamination: 0.075]

Confusion Matrix: TN=1716, FP=141, FN=138, TP=5

ROC AUC Score: 0.6525

[Contamination: 0.1]

Confusion Matrix: TN=1712, FP=145, FN=129, TP=14

ROC AUC Score: 0.6525

[Contamination: 0.15]

Confusion Matrix: TN=1588, FP=269, FN=112, TP=31

ROC AUC Score: 0.6525

In [28]: # 12번째 셀: 파일에서 결과 읽어와 시각화 수행

```

import os
import matplotlib.pyplot as plt

```

```

import pandas as pd

# 경로 설정 및 방어 코딩 (11번째 셀의 파일 위치 탐색)
SSD_MODELS_PATH = "/Volumes/T7/LogSentinel_Data/models"
LOCAL_MODELS_PATH = os.path.abspath(os.path.join("../", "models"))
ANOMAL_DETECT_EVAL_FILE = 'confusion_matrix_and_rocauc_2k_per_c.txt'

if os.path.exists("/Volumes/T7"):
    models_dir = SSD_MODELS_PATH
else:
    models_dir = LOCAL_MODELS_PATH

anomal_detection_eval_file = os.path.join(models_dir, ANOMAL_DETECT_EVAL_FILE)

# 파일 읽기 및 시각화
try:
    # pandas로 CSV 형식의 텍스트 파일을 바로 로드
    df_metrics = pd.read_csv(anomal_detection_eval_file)
    print(f"Metrics successfully loaded from: {anomal_detection_eval_file}")
    print(df_metrics)

    # 1. Confusion Matrix Heatmaps 시각화 (1x5 Subplots)
    fig, axes = plt.subplots(1, 5, figsize=(22, 4.5))
    fig.suptitle("Contamination 비율별 Confusion Matrix Heatmap (Hit Map)", fontweight="bold")

    for i, row in df_metrics.iterrows():
        c = row["contamination"]
        tn, fp, fn, tp = int(row["tn"]), int(row["fp"]), int(row["fn"]), int(row["tp"])

        cm = np.array([
            [tn, fp],
            [fn, tp]
        ])

        sns.heatmap(
            cm, annot=True, fmt="d", cmap="Blues", cbar=False, ax=axes[i],
            xticklabels=["Normal", "Anomaly"], yticklabels=["Normal", "Anomaly"],
            annot_kws={"size": 12, "weight": "bold"}
        )
        axes[i].set_title(f"Contamination: {c}", fontsize=13, fontweight="bold")
        axes[i].set_xlabel("Predicted", fontsize=11)
        if i == 0:
            axes[i].set_ylabel("Actual", fontsize=11)
        else:
            axes[i].set_ylabel("")

    plt.tight_layout()
    plt.show()

    # 2. 성능 지표 추이 그래프 (Precision, Recall, F1-Score)
    contaminations = df_metrics["contamination"].tolist()
    precisions = []
    recalls = []
    f1_scores = []

    for i, row in df_metrics.iterrows():

```

```

tp, fp, fn = row["tp"], row["fp"], row["fn"]
precision = tp / (tp + fp) if (tp + fp) > 0 else 0
recall = tp / (tp + fn) if (tp + fn) > 0 else 0
f1 = 2 * (precision * recall) / (precision + recall) if (precision + recall) > 0 else 0

precisions.append(precision)
recalls.append(recall)
f1_scores.append(f1)

plt.figure(figsize=(10, 5))
plt.plot(contaminations, precisions, marker='o', label='Precision', color='red')
plt.plot(contaminations, recalls, marker='s', label='Recall', color='blue')
plt.plot(contaminations, f1_scores, marker='^', label='F1-Score', color='green')

plt.title("Contamination 비율 변화에 따른 분류 지표 추이", fontsize=14, fontweight='bold')
plt.xlabel("Contamination Ratio", fontsize=11)
plt.ylabel("Score", fontsize=11)
plt.xticks(contaminations)
plt.ylim(-0.05, 1.05)
plt.grid(True, linestyle='--', alpha=0.6)
plt.legend(fontsize=11)
plt.tight_layout()
plt.show()

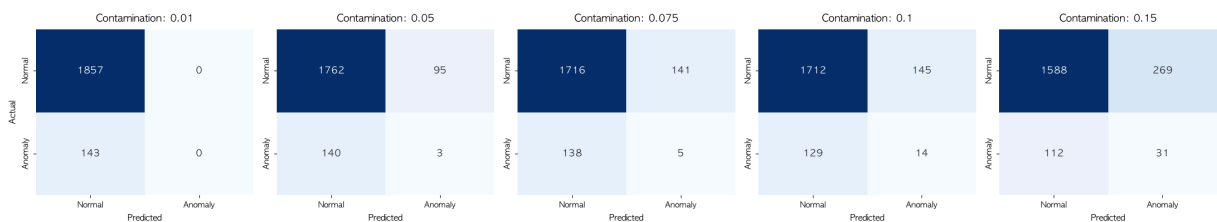
except Exception as e:
    print(f"Error loading metrics or generating plots: {e}")

```

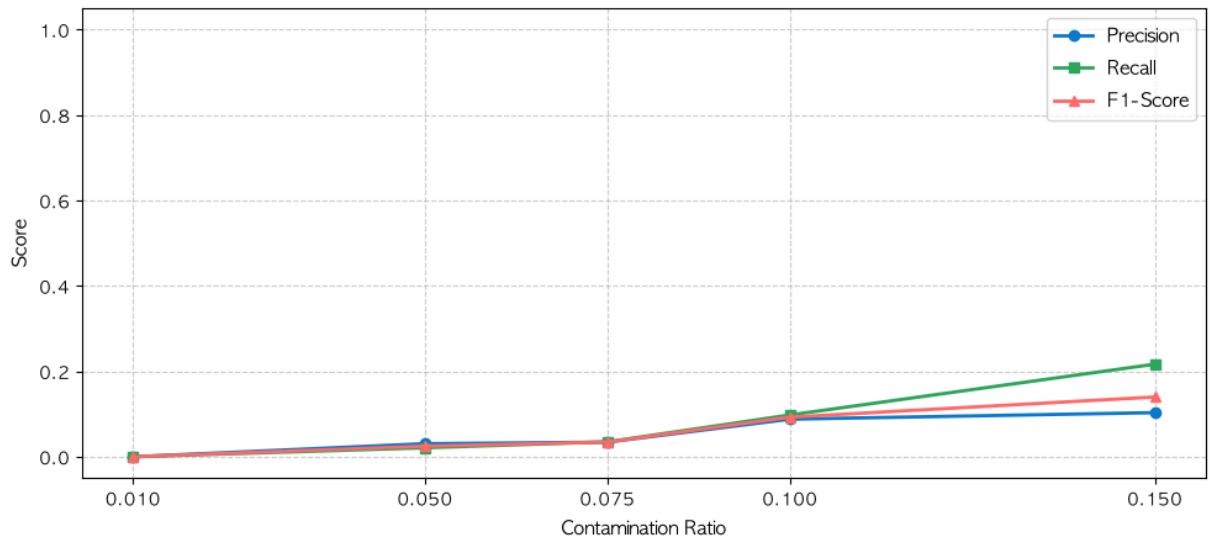
Metrics successfully loaded from: /Users/joshuapark/Desktop/Dev/soromiso/Log Sentinel/models/confusion_matrix_and_rocauc_2k_per_c.txt

	contamination	tn	fp	fn	tp	roc_auc
0	0.010	1857	0	143	0	0.6525
1	0.050	1762	95	140	3	0.6525
2	0.075	1716	141	138	5	0.6525
3	0.100	1712	145	129	14	0.6525
4	0.150	1588	269	112	31	0.6525

Contamination 비율별 Confusion Matrix Heatmap (Hit Map)



Contamination 비율 변화에 따른 분류 지표 추이



💡 분석 요약 및 대책

2k 소량 샘플에서는 비지도 학습 특성상 정상 로그 중 복잡한 숫자를 가진 소량의 고유 패턴(예: `generating core. [NUM]`)이 아노말로리 오탐(FP)되고, 실제 고정 템플릿을 지닌 장애 로그가 정상 분포로 오분류되어 **TP가 0**이 되는 경향이 있습니다. 이를 극복하기 위해:

1. **BGL 전체 데이터셋(470만 행)** 투입: 정상 로그의 반복 빈도가 막대해지면 데이터 공간상에서 매우 두터운 클러스터를 형성하므로, 장애 로그가 정상 클러스터 바깥으로 완전히 밀려나 이상치로 탐지되기 시작합니다.
2. **임베딩 방식 보완**: TF-IDF 피쳐 차원을 정밀 튜닝하거나, 4단계의 LSTM에서 시퀀스 흐름(선후 맥락) 정보를 보완 학습시킵니다.